INCH-POUND

MIL-DTL-17/158B
11 February 2005
SUPERSEDING
MIL-C-17/158A
18 July 1985

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/158-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-17.

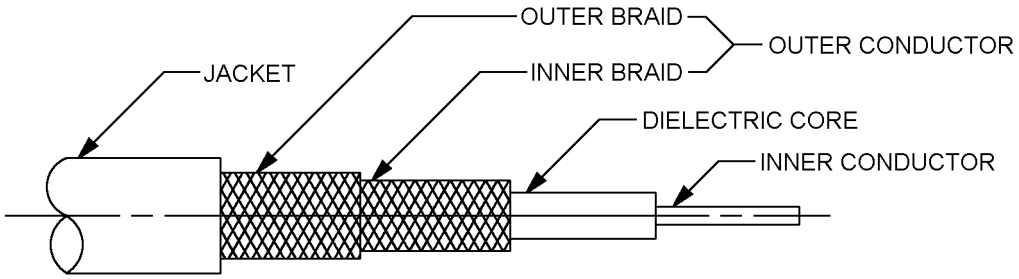


FIGURE 1. Configuration.

AMSC N/A

FSC 6145

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TABLE I. Description.

Component	Construction details
Inner conductor	Solid silver-coated, copper-covered steel wire. Overall diameter: 0.037 inch ± 0.001
Dielectric core	Type F-1: Solid extruded PTFE Diameter: 0.116 Inch ± .005
Outer conductor	Double braid of AWG No. 36 silver-coated copper wire. Diameter: 0.171 inch maximum
Inner braid	Coverage: 94.8% nominal Carriers: 16 Ends: 7 Picks/inch: 11.5 ± 10%
Outer braid	Coverage: 93.1% nominal Carriers: 16 Ends: 7 Picks/inch: 14.5 ± 10%
Jacket	Type IX: FEP Diameter: 0.195 inch ± 0.005

CAUTION IS DIRECTED TO THE APPLICATION OF THIS CABLE ABOVE 400 MHz. ATTENUATION IS TESTED ONLY AT 400 MHz. SRL AND POWER HANDLING CAPABILITIES ARE NOT STIPULATED HEREIN.

ENGINEERING INFORMATION

Continuous working voltage: 1,400 V rms, maximum.

Velocity of propagation: 69.5 percent, nominal.

Operating temperature range: -55°C to +200°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 1.95 ohms per 100 feet.

Elongation: 1 percent, minimum.

Tensile strength: 110 $klb_f/inch^2$, minimum.

Engineering note: This cable is useful in general purpose, high temperature applications. (See connector series “TNC”, “BNC”, and “SMA” in accordance with MIL-PRF-39012.)

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REQUIREMENTS

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 4 pounds, minimum; 15 pounds, maximum.

Aging stability: Not applicable.

Stress crack resistance: +230°C ± 5°C; mandrel size 5 times the jacket diameter.

Outer conductor integrity: Not applicable.

Cold bend: -55°C ± 2°C.

Dimensional stability: +200°C ± 5°C.

Inner conductor from core: 0.250 inch, maximum.

Inner conductor from jacket: 0.312 inch, maximum.

Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Applicable.

Weight: 0.056 pound per foot, maximum.

Electrical:

Continuity: Applicable.

Spark test: 2,000 V rms, +10%, -0%.

Voltage withstanding: 5,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ohms ± 2.

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Attenuation: 11.7 dB/100 ft maximum at 400 MHz.

Structural return loss: Not applicable.

Capacitance: 32 pF per foot, maximum.

Capacitance, stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part or Identifying Number (PIN): See table II.

Supersession data: See table II.

TABLE II. Cross-reference of PIN.

PIN	Superseded PIN or type designation
M17/158-00001	RG-142B/U in accordance with MIL-C-17

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-C-17, this document references the following:

MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA - CC

Preparing activity:
DLA – CC

(Project 6145-2381-003)

Review activities:
Army - AR, AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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